

OPERATING MANUAL
for
Magnex Model E7006
Emergency Discharge Unit

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MAGNEX SCIENTIFIC LIMITED

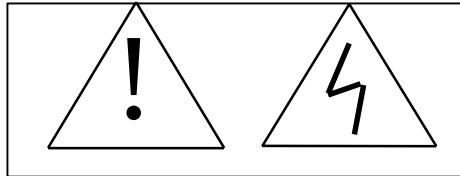
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CONTENTS

1.	SAFETY	3
2.	IMPORTANT NOTICE	4
3.	SPECIFICATIONS	6
4.	DESCRIPTION AND OPERATING PRINCIPLE	7
5.	INSTALLATION	9
6.	SYSTEM INTERCONNECTING DIAGRAM	11
7.	PERIODIC TESTING	14
8.	TROUBLE-SHOOTING	16
9.	USER SERVICEABLE PARTS	17
10.	SETTING THE BO DUMP CLOCK	18
11.	BATTERY CARE	20

1. SAFETY



1. WARNING - Do not remove any protective covers whilst this unit is open.
LIVE PARTS INSIDE.
2. WARNING - For continued protection against risk of fire or electric shock, replace fuses with the same type, rating and UL listing.
3. CAUTION - This unit contains internal protection fuses. Follow the guidelines in Trouble-shooting section before replacement of these devices.
4. CAUTION - This unit has been pre-set to a voltage range suitable for the national supply. The unit has a selectable input voltage, 115V~ or 230V~. Fuses F1 and F2 must be rated to suit the input voltage, if selected to 115V~ they must be rated at T250mA 250V. If selected to 230V~ they must be rated at T125mA 250V.

To select the mains supply voltage or replace mains fuses, prise open the flap on the mains input socket using a suitable tool in the slot provided. This affords access to the voltage select drum and two fuse trays.

To select the supply voltage, pull out the drum and rotate until the 230V~ or 115V~ lettering is uppermost, ie. in line with the drum slots, and replace. When the socket flap is replaced the correct voltage must be visible through the window.

To replace the input fuse, pull out the fuse trays and replace with the correct value of fuse. Re-insert the fuse trays, noting the correct orientation of the arrows.

5. WARNING - If replacing the back-up batteries, follow the instructions in the Battery Care section of this manual.
6. WARNING - The batteries and fuses are the only serviceable parts within this unit. No other circuitry should be tampered with as this may effect the operation of the discharge system.

2. IMPORTANT NOTICE

Please inspect goods immediately on arrival for possible transit damage and notify Magnex Scientific Limited within 3 days of receipt of goods.

Failure to do this will invalidate any possible claim.

DISCLAIMER

Every precaution has been taken in the preparation of this publication. Magnex Scientific Limited assumes no responsibility for errors or omissions. Neither is any liability assumed for damages resulting from the use of the information contained herein.

Details of circuitry and user serviceable spare parts are available upon request from :-

Magnex Scientific Limited
The Magnet Technology Centre
6 Mead Road
Oxford Industrial Park
Yarnton
Oxford OX5 1 QU, UK

CE NOTICE

Marking by the symbol **CE** indicates compliance of this device to the EMC (electromagnetic compatibility) and LV (low voltage) directives of the European Community. This unit is to be installed and operated as detailed. Any modification or maintenance procedure undertaken which is not approved by Magnex Scientific Ltd could nullify the **CE** marking of this product and lead to prosecution. A 'Declaration Of Conformity' in accordance with the above directives has been made and is located at Magnex Scientific, Yarnton, Oxfordshire, UK.

3. SPECIFICATIONS

Suitable quench heater and connecting cable resistance	:	6.8 Ω minimum 18.0 Ω maximum
Quench voltage	:	18V $\overline{\text{---}}$ maximum
Quench cycle duration	:	30 seconds (+/- 20%)
Test circuit control input	:	TTL/CMOS Compatible
Test circuit monitor output	:	TTL/CMOS Compatible
B0 quench heater current	:	19 to 100 mA (variable)
Mains voltage	:	115/ 230 V $\sim$ selectable
Mains input current	:	250 mA @ 115V $\sim$ 125 mA @ 230V $\sim$
Mains input frequency	:	50/60 Hz
Fuse ratings	:	F1, F2 T125 mA for 230V $\sim$ supply OR T250 mA for 115V $\sim$ supply F3, F4 T1.6A 250V
Dimensions		
Height	:	230mm
Width	:	230mm
Depth	:	125mm

4. DESCRIPTION AND OPERATING PRINCIPLE

Magnex Model E7006 Emergency Discharge Unit

Features

- * Magnet quench indication
- * Indication that the discharge button has been operated
- * Remote emergency discharge button option
- * Battery backed
- * Heater and battery test facility
- * Programmable B0 coil dump clock timer.

Description

The principle function of the E7006 emergency discharge unit (EDU) is to initiate a 'quench' in order to run down a superconducting magnet in case of any emergency. This is achieved by passing current through a heater built into the magnet windings. A mains derived dc voltage drives the current into the heater. The unit is also battery backed to ensure operation during a power failure.

A test facility is provided to check the condition of the battery and the integrity of the heater circuit.

A further facility is a clock controlled output current used to energize the B0 shield switch heater allowing the fluxes within the B0 coil to decay.

Operation

To initiate a magnet quench; lift the emergency discharge button guard and press the red button. Current will flow into the quench heater causing the magnet to quench very quickly. Quench current will flow for approximately 30 seconds. This will be indicated by the illumination of the front panel magnet rundown button.

WARNING - Following a quench, the bursting disc must be replaced and care must be taken to ensure that the cryostat is properly sealed to prevent air entering the helium reservoir.

The condition of the battery and heater circuit is observed by pressing the **TEST** button. The **BATTERY OK** and **HEATER OK** LED's will only illuminate if the battery capacity is okay and the quench heater circuit is intact.

The test circuit can also be initiated by applying a TTL input signal via the 9 'D' **PSU** socket. A TTL output signal is available to indicate heater integrity, making it suitable for inclusion in digital control systems.

The B0 dump clock timer is programmed to pass current through the B0 shield switch heater for 1 minute every 24 hours. (This must be set during installation when the clock is also set to the correct time).

This setting can be manually over-ridden by pressing the **CHANGE** button or via an external TTL signal applied to the **PSU** or **TEST** connectors.

The **B0 DUMP ON** LED will illuminate to indicate that the B0 shield switch heater is energised.

CAUTION - When the B0 shield switch heater is quenched by these methods, the winding remains permanently 'dumped' until the **CHANGE** button is pressed or the TTL signal is removed.

5. INSTALLATION

The EDU and remote discharge buttons are wall mounted using the fixings provided.

To install follow the instructions below :

- 1) Mark and drill the wall locations using the mounting templates supplied.
- 2) Insert the fixing lugs into the holes. Remove the lid of the button box; this allows access to the box fixing holes.
- 3) Offer the unit/buttons up to the wall holes and insert the fixing screws, tighten thoroughly. Replace the button box lids.
- 4) Ensure that the unit has been set to the correct mains input voltage. Connect mains power and ensure the power led illuminates.
- 5) Fit cable ferrite to cable C0091A12.5 20mm from 25 'D' plug end. Secure using cable tie.
- 6) Connect the cables as shown on the interconnect diagram.
- 7) Unlock door. Reconnect battery terminal 3.

CAUTION - Failure to reconnect the battery will mean that the magnet 'quench', B0 dump circuit and the B0 dump clock will not function in the event of a mains power failure.

- 8) Set the B0 dump clock (refer to setting the B0 dump clock section).

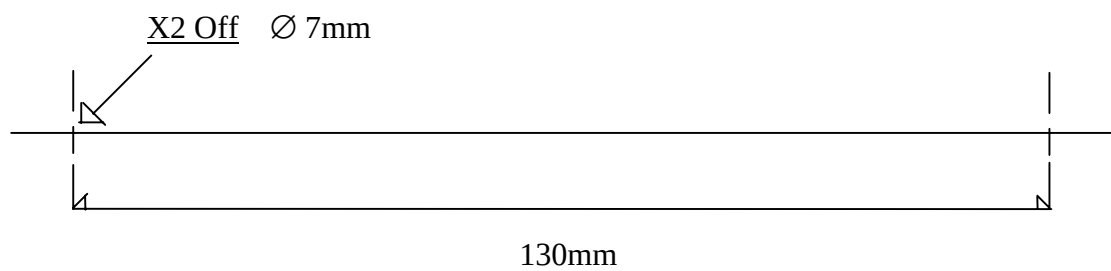
CAUTION - Failure to set the clock may cause the B0 dump function to remain energised.

- 9) Press the RESET button and ensure that the PCB LED is extinguished. Relock the door and remove key.
- 10) Press the test button and ensure that both the BATTERY OK and heater OK led's illuminate (The BATTERY OK led may not illuminate if the batteries have not been charged for at least 24 hours).

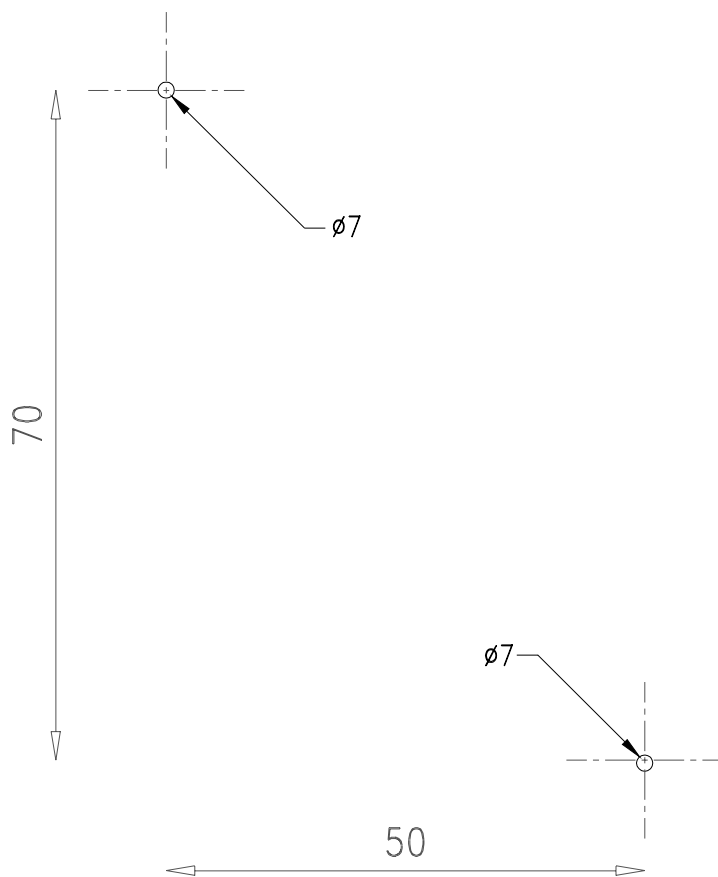
Note

Once the magnet is operational, cables C0008 and C0021 may be removed.

E7006 EMERGENCY DISCHARGE UNIT MOUNTING TEMPLATE

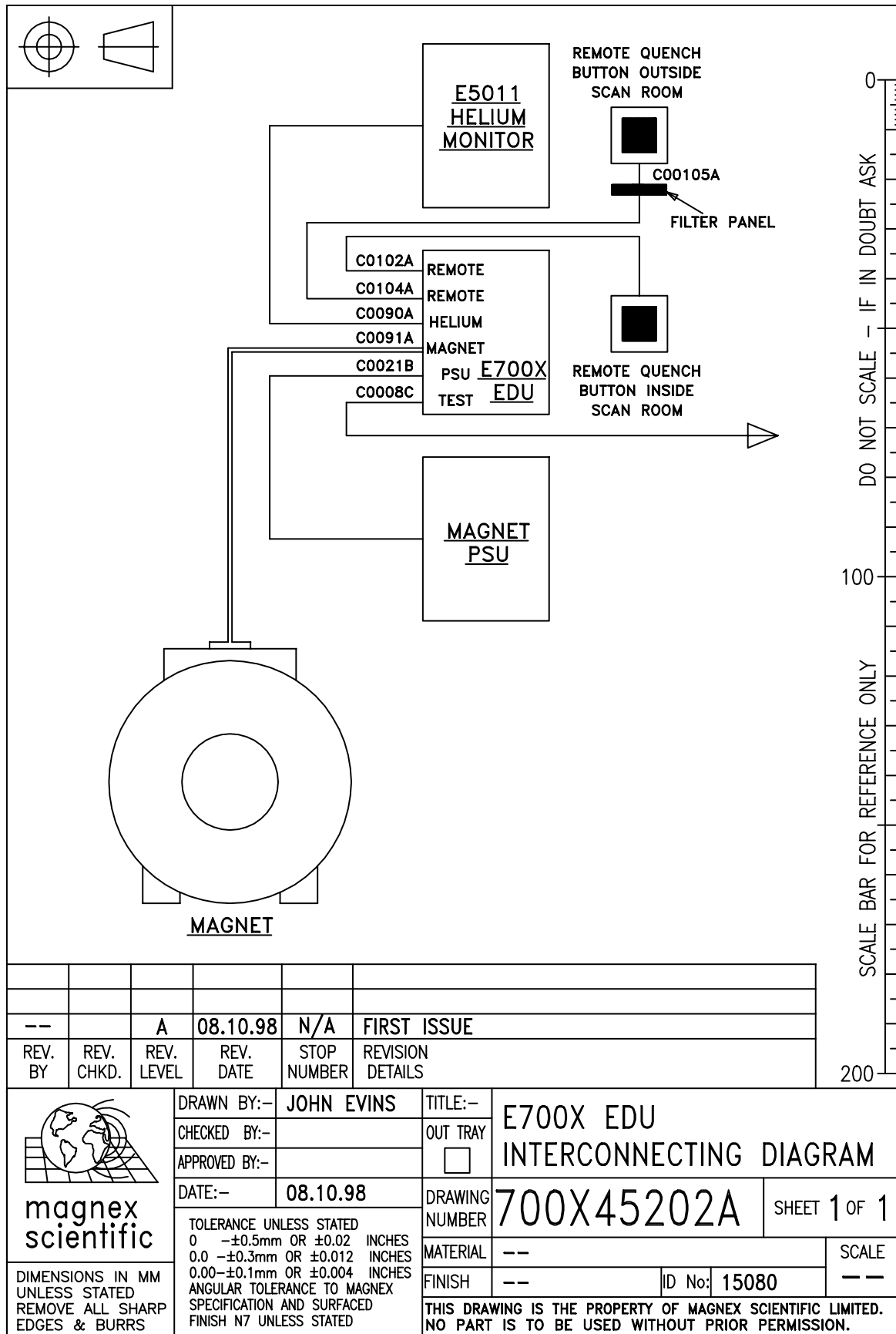


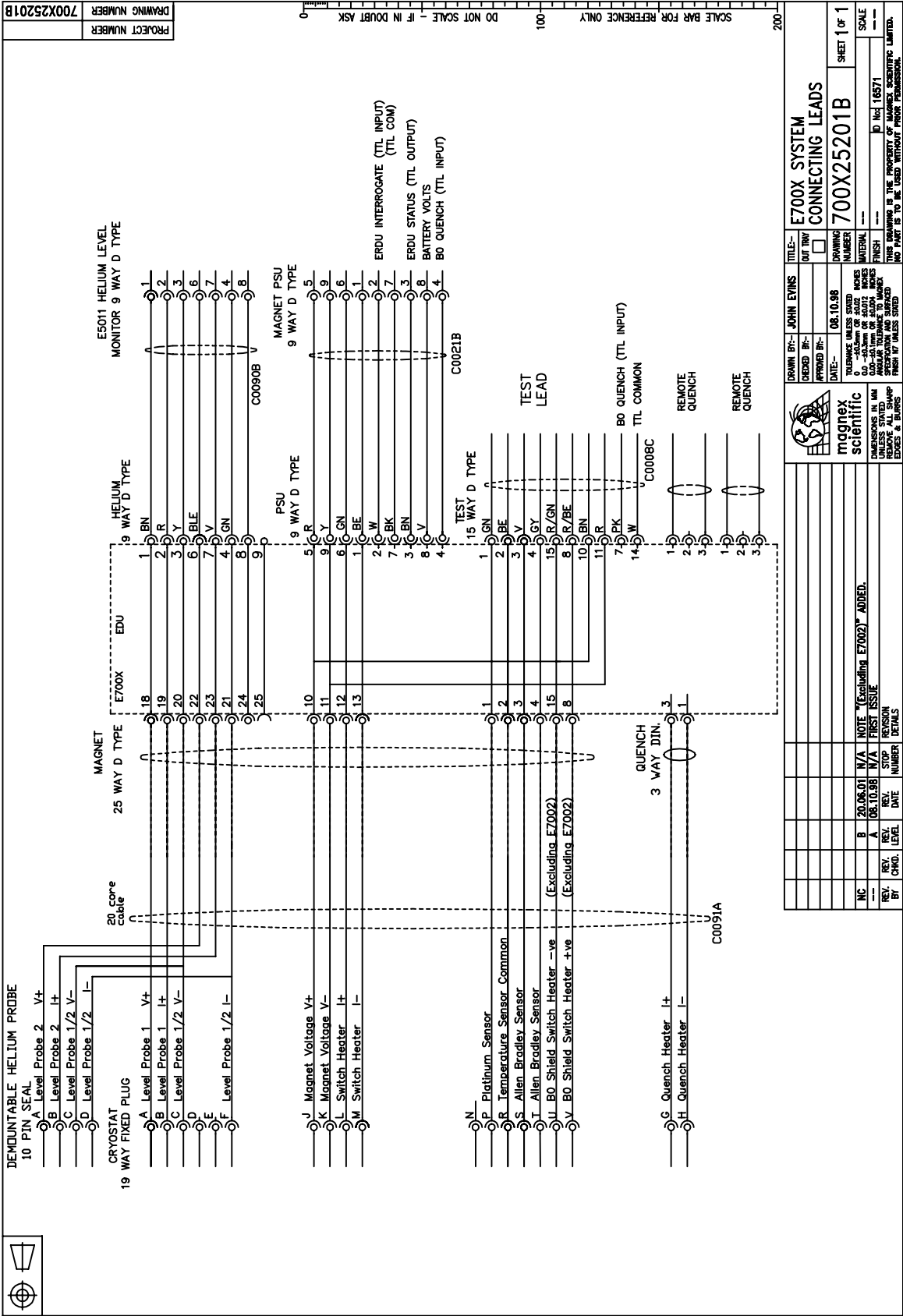
REMOTE DISCHARGE BUTTON MOUNTING TEMPLATE



6. SYSTEM INTERCONNECTING DIAGRAMS

BLANK



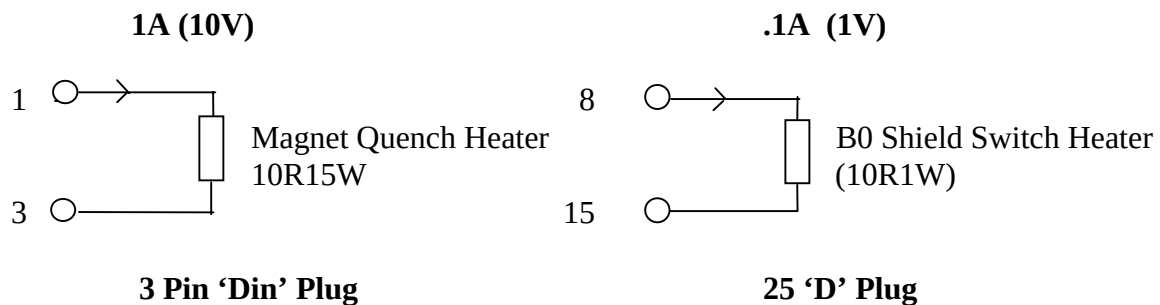


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7. PERIODIC TESTING

It is recommended that the quench and B0 dump currents are periodically tested to ensure correct operation.

The following test loads must be attached in place of the service cable.



Magnet Quench Heater Current

To test the magnet quench heater current, press the emergency discharge button on the front panel. The button will illuminate. A minimum of 10V must be developed across the 'magnet quench heater' load indicating that 1A of current is flowing. This current will flow for approximately 30 seconds then cease.

WARNING : Do not reconnect the service cable during this time as a magnet quench will result.

The remote quench buttons must also be tested, they are tested in the same manner, however the buttons will not illuminate to indicate that current is flowing.

B0 Dump Switch Heater Current

To test the B0 dump switch heater current, press the clock timer **CHANGE** button. The **B0 DUMP ON** LED will illuminate and a minimum of 1V must be developed across the 'B0 shield switch heater' load indicating that 100mA of current is flowing.

The current can be adjusted using VR2 mounted on the PCB. Press the **CHANGE** button to stop the flow of current.

Check the clock timer settings to ensure that the time is correct and the B0 shield switch heater is energised for 1 minute every 24 hours. To review the set programmes press and hold the **PROGRAM** button.

Ensure that the slider switch is in the **TIMED** position.

Check Batteries

Check the dated 'tested' label on the batteries to ensure that they are within their service life (refer to battery care section).

Completion

Relock door and remove key. Reconnect service cable. Press **TEST** button. The **HEATER OK** and **BATTERY OK** LED's must illuminate.

8. TROUBLE SHOOTING

Front panel display conditions can give an indication of the problem.

<u>Display</u>	<u>Possible Fault</u>
No POWER LED	Loss of F1, F2 (rear panel mounted) or F3 (PCB mounted). *
HEATER OK LED not illuminated when TEST button pressed.	Loss of F4 (PCB mounted) *. Heater and cable resistance > 18Ω. Heater open circuit.
BATTERY OK LED not illuminated when TEST button pressed.	Batteries not fully charged or require replacing **(ensure batteries have charged for at least 24 hours prior to testing)
B0 DUMP ON LED not illuminated when B0 dump circuit operated	B0 shield switch heater open circuit.
B0 DUMP ON LED permanently illuminated	B0 dump clock timer incorrectly set (refer to setting B0 dump clock section)
B0 DUMP CLOCK MODULE Display blank on initial power up	Internal battery needs to charge for 30 minutes (refer to setting B0 dump clock section)

For any faults first check that proper connection is made both at the cryostat and at the unit.

If these tests do not highlight the fault then check the continuity and resistances at the cryostat end.

* To change the PCB mounted fuses follow the instructions given below :-

1. Remove mains supply lead.
2. Unlock door.
3. Locate fuse.
4. Remove fuse insulation cover, remove 'blown' fuse and replace with fuse of correct ratings.
5. Replace fuse insulation cover, relock door and remove key.

** Refer to Battery Care Section

9. USER SERVICEABLE PARTS

<u>COMPONENT</u>		<u>DESCRIPTION</u>	<u>PART NUMBER</u>
B1		12V 1.2 A/h Battery	P188160001
B2		6 V 1.2 A/h Battery	P188160002
F1, F2	OR	5 X 20mm T125mA 250V Fuse	P188030005
		5 X 20mm T250mA 250V Fuse	P188030007
F3, F4		5 X 20mm T1.6A 250V Fuse	P188030012

10. SETTING THE B0 DUMP CLOCK

Clock Timer

The clock timer is used to energise the B0 switch heater allowing the fluxes within the B0 coil to decay.

It is recommended that the B0 switch heater is energised for 1 minute every 24 hours.

CAUTION - Failure to set the clock timer may cause the B0 coil to be permanently energised.

CAUTION - The clock timer is mains operated and battery backed, however it is recommended that the clock settings be checked should the unit be disconnected from the mains supply for periods greater than 72 hours.

Clock Setting

The clock timer operates in 24 hour mode and has the option to set up to 4 individual programmes.

A programme is a pair of ON/OFF settings which will dictate when the appliance will switch ON & OFF.

To prepare the module for programming set the slider switch to the **TIMED** position. Press the **CHANGE** and **PROGRAM** buttons simultaneously until the display goes blank.

The display will reappear to show a clock and flashing hours digits. The module is now ready to enter the correct local time.

Press the **CHANGE** button to advance the hour setting. Press the **PROGRAM** button to select the minute digits, they should start flashing. Use the **CHANGE** button to advance the minute setting. Press the **PROGRAM** button once to store the clock time.

The display should now show --:-- on with a flashing on. Set the first time on time to 00.01 as before. This has set the timer to switch on at 12 o'clock midnight and switch off at 1 minute past midnight. Programmes 2, 3 and 4 can be ignored, by pressing the **CHANGE** until the display shows the normal operating mode (current time).

Instant ON/OFF (over-ride)

To over-ride the programme in operation and to immediately switch ON or OFF press the **CHANGE** button.

Note

Button pauses greater than 1 minute during programming will result in automatic return to the normal operating mode.

On, Off and Timed slider switch

The **TIMED** position allows the module to operate using the programmed settings. The module must be left in this **MODE** to ensure that the B0 coil is dumped at regular intervals. The ON position permanently dumps the B0 coil, regardless of the timer settings. The OFF position disables the timer settings. The clock continues to function as normal.

11. BATTERY CARE

Manufacturers data shows that the batteries have a service life of between 4 ½ and 5 ½ years.

It is recommended that the batteries are replaced after the minimum time has elapsed.

Refer to dated 'tested' label on the batteries.

Replacement Batteries

Replacement battery packs can be obtained from Magnex Scientific Limited.

To replace batteries :-

1. Unlock door.
2. Disconnect battery terminals. Remove battery clamp by removing fixing screws. Lift out old batteries.
3. Fix new batteries to base plate using double sided tape provided.
4. Replace battery clamp firmly.
5. Reconnect battery terminals (leads and terminals are labelled). OBSERVE CORRECT POLARITY.
6. Relock door and remove key.
7. Ensure batteries are charged for at least 24 hours prior to testing.

Magnex Ref : QA2
Rev : 0

Dear Customer,

In order to help us maintain our quality records and ensure traceability of our products, could you please complete and return this form if this unit was, or is to be, supplied by you with a system different to the one listed.

Unit Description

Unit Serial Number

Project Number Supply by Magnex on

System Description

Please complete the duplicate form overleaf if necessary and return to :-

The Quality Assurance Manager
The Magnet Technology Centre
6 Mead Road
Oxford Industrial Park
Yarnton
Oxford OX5 1QU, UK

Your co-operation is appreciated.